Fraction division and reciprocals

Explore and build the idea

20 daily questions include C1 and C2. Use separate paper when helpful.

Lesson focus: I can connect the reciprocal rule to equal groups and use it in measurement problems.

R1. Find $\frac{1}{2} + \frac{1}{3}$.

R2. Complete the ratio-table pair: (3, 4), (9, __).

R3. 9 of 12 tiles are green. What percent are green?

R4. How many $\frac{1}{4}$-meter pieces fit in 3 meters?

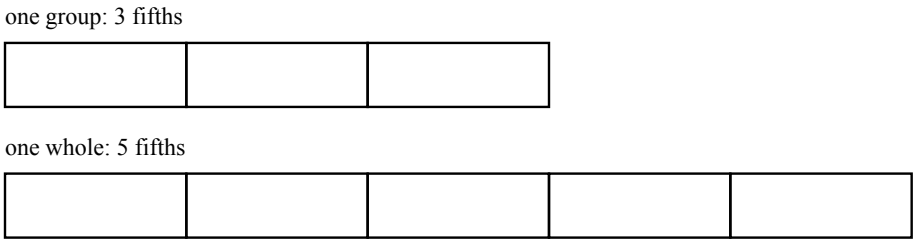
P1. Share $\frac{3}{4}$ liter equally among 5 bottles. How much per bottle?

P2. How many $\frac{3}{5}$-cup groups fit in 4 cups? Include a fraction of a group.

P3. Find $\frac{7}{8} \div \frac{1}{4}$. Check by multiplying the quotient by $\frac{1}{4}$.

P4. Find $1\frac{7}{8} \div \frac{5}{8}$.

Lesson model



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Practice and compare methods

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P5. A cord is $3\frac{1}{2}$ meters long. Each piece is $\frac{7}{8}$ meter. How many pieces?

P6. A rectangle has area $\frac{5}{6}$ square meter and width $\frac{2}{3}$ meter. Find its length.

P7. A box has edges $\frac{2}{3}$, $\frac{3}{4}$, and 2 meters. Find its volume.

F. Complete: $\frac{4}{5} \div \frac{2}{3} = \frac{4}{5} \times __ = __$. Check by multiplying by $\frac{2}{3}$.

D. Zoe writes $\frac{7}{8} \div \frac{1}{4} = \frac{8}{7} \times \frac{1}{4} = \frac{2}{7}$. Explain which number should be inverted.

M1. Find $3 \times \frac{2}{3}$.

M2. A recipe uses 2 cups beans for 3 cups rice. How many cups beans go with $\frac{9}{2}$ cups rice?

M3. 3 markers cost $1.95. Find dollars per marker.

Worked example - compare with the slide model

Find $\frac{2}{3} \div \frac{3}{5}$.

One whole contains $\frac{5}{3}$ groups of size $\frac{3}{5}$.

Two thirds of a whole contains $\frac{2}{3} \times \frac{5}{3}$ groups.

Multiply to get $\frac{10}{9}$, or $1\frac{1}{9}$.

Result: $\mathbf{1\frac{1}{9}}$

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Challenge, explain and check

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C1. A rectangular card has area $\frac{5}{6}$ square meter and width $\frac{2}{3}$ meter. Find its length, then the distance around its edge.

C2. A box measures $\frac{3}{4}$ meter by $\frac{2}{3}$ meter by $2\ \frac{1}{2}$ meters. Find its volume. Two such boxes occupy a 3-cubic-meter space; how much volume is unoccupied?

Finish the lesson

S. I can check a fraction quotient by multiplying it by ___.

E. A ribbon is $2\ \frac{5}{8}$ yards long. Each piece is $\frac{3}{8}$ yard. How many pieces can be cut?